

Ken ZOU

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RESEARCH EXPERIENCE

University of California, Berkeley

Apr. 2026 – Present

Visiting Student Researcher – Zuerch Lab

Berkeley, CA

- Advisor: Prof. Michael Werner Zuerch | mwz@berkeley.edu
- Contributed to ongoing experiments in ultrafast X-ray spectroscopy and solid-state high harmonic generation (sHHG).
- Operated laser and detection setups. Prepared and characterized samples, performed data processing and analysis.

ESPCI Paris – PSL (PSL University)

Jan. 2026 – Apr. 2026

Research Intern – LPEM Research Group

Paris, France

- Advisor: Prof. Cheryl Feuillet-Palma | cheryl.feuilletpalma@espci.fr
- Investigated the competition between magnetic impurities and superconducting order in Nb thin films with controlled Manganese ion implantation doses. Analyzed the influence of controlled disorder geometry.

NTT Basic Research Laboratories (NTT BRL)

July 2025 – Dec. 2025

R&D Intern – Nanodevices Research Group

Atsugi, Japan

- Advisor: Dr. Gento Yamahata | gento.yamahata@ntt.com
- Characterised silicon-based single-electron and single-hole pumps at cryogenic temperatures (dilution refrigerator) for qubit devices and metrology applications. Benchmarked electron vs. hole performance in a single quantum dot.

Sorbonne University – French Chemistry Tournament (TFChim)

Feb. 2024 – June 2025

Research Team Member – ESPCI Team

Paris, France

- Advisor: Prof. Yvette Tran | yvette.tran@espci.fr
- Member of the back-to-back winning ESPCI team (2024 and 2025) in a highly competitive national competition against ENS, Polytechnique, and Sorbonne, involving oral defence of original research in chemistry.
- Led synthesis of zinc dendritic structures by electrochemical growth and organic synthesis of fractal benzenes.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Starting in Sept. 2026

M.Sc. in Quantum Science and Engineering

Lausanne, Switzerland

- Admitted with the EPFL Master Excellence Fellowship (awarded to the top 3% of admitted students).
- Coursework: Quantum Computing, Quantum Physics, Computational Methods for Molecules, Quantum Simulations of Materials, Statistical Mechanics, Molecular Quantum Dynamics, Machine Learning for Physicists.

ESPCI Paris – PSL (PSL University)

Sept. 2023 – Present

M.Sc. in Physics & Chemistry

Paris, France

- GPA: 4.0/4.0 | 1st in cohort (1/88, 2024), 2nd in cohort (2/106, 2025) | Highest Honors
- Specialization: Waves and Light-Matter Interactions, Relativistic and Quantum Engineering, Magnetism and Superconductivity, Advanced Condensed Matter Physics, Synthetic Chemistry.

Lycée Louis-Le-Grand (LLG)

Sept. 2021 – July 2023

Classes Préparatoires (CPGE), Maths-Physics-Chemistry track

Paris, France

- GPA: 4.0/4.0 | Top 5% across national candidates.
- Advanced oral and written coursework problem-solving in pure & applied mathematics, theoretical & experimental physics and chemistry, and computer science in the best French university-level formation, sending the highest proportion of students to top French engineering schools (*Grandes Écoles*).
- Eligible for École Polytechnique, ENS Paris – PSL, Mines Paris – PSL through highly competitive examinations.

WORK EXPERIENCE

Commission des Titres d'Ingénieur (CTI)

Oct. 2024 – Present

Engineering Student Expert

Paris, France

- Audited French engineering schools alongside domain experts; prepared evaluation frameworks and contributed to official accreditation reports (Bologna Process compliance).

French Ministry of Education

Sept. 2024 – June 2025

Oral Examiner in Physics & Chemistry – LLG & ENCPB (Pierre-Gilles de Gennes)

Paris, France

- Assessed 140+ students in real-exam oral conditions (6 h/week) within France's most selective pre-engineering courses; provided targeted scientific feedback.

Physique-Chimie Avenir (PCA) – ESPCI Junior Enterprise

Mar. 2024 – Mar. 2025

Quality Manager

Paris, France

- Managed quality assurance across 20 client studies (Bouygues, Solvay, EDF, Veolia; 50 k€ revenue). Implemented document management and archiving workflows in line with CNJE standards.
- Served as a board member and contributed to strategic decision-making with the presidency's action plan.

AWARDS & FELLOWSHIPS

ESPCI Alumni – Nominee for the 2026 Best Industrial Internship Award.

EPFL Master Excellence Fellowship – 20 000 CHF/year and guaranteed housing (2026-2028).

French Chemistry Tournament (TFChim 2025) – Winner.

Joliot ESPCI Scholarship – 3 071€ (2023-2025), 1 535€ (2026).

French Government CROUS Scholarship – 2597€ (2021), 2701€ (2022), 3071€ (2023-2024), 2163€ (2025-2026).

French Government Merit-Based Baccalaureate Grant – 900€ (2021).

LEADERSHIP & COMMUNITY ENGAGEMENT

Terraskola & Acadomia – Private oral examiner & Undergraduate-level Tutor in Physics, Chemistry. [Feb. 2026 – Present]

ESPCI's Cohort 142 Student Council – Representative. [Nov. 2023 – Present]

Prép'Assist – Undergraduate-level Tutor in Maths, Physics, Chemistry. [Sept. 2023 – June 2025]

Forum Horizon Chimie (FHC) 2024, 38th Edition – Digital Manager. [Jan. 2024 – Oct. 2024]

ESPCI first-year pre-entry course in Organic Chemistry – 15-hour course Teaching Assistant. [Aug. 2024]

Ambition Science ParisTech ENCPB (Climbing Together to Success) – Undergraduate-level Tutor in Maths, Physics & Chemistry. Promoted social diversity and inclusion by actively supporting students from underrepresented backgrounds and helping them navigate through academic and administrative challenges. [Oct. 2023 – June 2024]

CERTIFICATIONS

TOEFL iBT – Score: 104/120. [Earned in Nov. 2025]

IBM Quantum – Basics in Quantum Information & Fundamentals of Quantum Algorithms. [Earned in Oct. 2025]

CNJE – Consulting Auditor Test Success. [Earned in Nov. 2024]

ENS Paris – PSL & PSL University – Basics in Audiology & Psychoacoustics. [Earned in Nov. 2024]

EdX Harvard – Basics in Neuroscience & Neuropsychology. [Earned in Aug. 2023]

Deutsches Sprachdiplom (DSD) – B1 German Proficiency Certification. [Earned in Mar. 2019]

SKILLS

Research skills: Trustworthy, organizational, result-oriented, adaptable, creative, independent.

IT skills: Python (NumPy, SciPy, QuTiP), MATLAB, SQL, Git, LaTeX, Excel, ImageJ, Klayout, Igor Pro, Quantum ESPRESSO, OCEAN, PySCF, Qiskit, Vesta.

Languages: French (Native), English (TOEFL iBT 104), German (DSD B1), Japanese & Cantonese (Basics).

Interests: Piano, Table Tennis, Athletics, Music, Mentorship, Philosophy, Neuroscience, Quantum Chemistry.